STAT3004

Stochastic Processes and Probability

A course book from Lectures 1–31



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About this book

This book develops the material in all 31 supplied STAT3004 lecture sets. Its order follows the course: stochastic-process models, discrete and continuous Markov chains, Poisson processes and queues, then the measure-theoretic foundations of probability and modes of convergence. Lecture 31 closes with a brief introduction to maxima. The final appendix maps every lecture to the relevant sections. All diagrams in this book are drawn in TikZ and are editable in the source.

The text expands the handwritten notes into self-contained explanations, derivations, worked examples and exercises. Statements are given with their assumptions; in particular, a stationary distribution need not be a limiting distribution, and convergence in distribution does not control expectations without additional conditions. Conditional expectation is only signposted in the final lecture, so it is outside the developed scope here.

How to use the book

Read a chapter's motivating model before its proofs. Work through its example with pencil and paper, then attempt the exercises without looking at the solutions. The final part supplies the probability language used implicitly in the earlier stochastic-process chapters; it can also be read first as a foundations course. A short notation guide appears below.

Notation

Symbol	Meaning
$\Pr(A)$, $\mathbb{E}X$, $\operatorname{Var}(X)$	probability, expectation, variance
X_n, $X(t)$	discrete-time and continuous-time processes
$\boldsymbol{P}$, $\boldsymbol{Q}$	transition matrix and generator
T_A, τ	hitting time and absorption time
π	stationary distribution
$X_n \xrightarrow{\text{a.s.},\Pr,L^p,d} X$	almost-sure, probability, L^p, distribution convergence

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Part I

Models in discrete time

Chapter 1

Stochastic processes and models

A stochastic process is a collection of random variables indexed by time or another parameter. The index tells us when to look; the random variable tells us what is observed. A random walk records position after each step, a branching process records population by generation, and a queue records customers in a system at each instant. These are different models because their state variables, clocks and dependence assumptions differ.

1.1 The object being modelled

Definition 1.1. A stochastic process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with state space S and index set T is a family $\{X_t : t \in T\}$ of S-valued random variables. If $T = \{0, 1, \ldots\}$, time is discrete; if $T = [0, \infty)$, it is continuous.

For one outcome ω, the function $t \mapsto X_t(\omega)$ is a *sample path*. The distribution of X_t at one time is a marginal distribution. The joint distributions of $(X_{t_1}, \ldots, X_{t_k})$ for every finite selection of times are the process's *finite-dimensional distributions*. Marginals alone do not specify dependence across time.

Worked example 1.2. Let B be a fair Bernoulli variable. Both $X_n = B$ for every n and Y_n consisting of independent fair Bernoulli variables have Bernoulli$(1/2)$ marginals. Yet $\Pr(X_1 = X_2) = 1$, while $\Pr(Y_1 = Y_2) = 1/2$. A model for repeated observations must say how they are coupled.

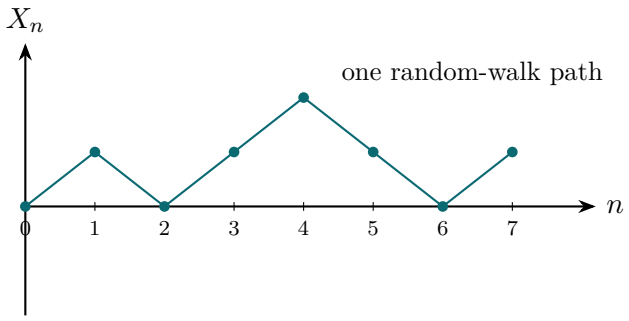


Figure 1.1: A sample path is one possible outcome of the entire process.

1.2 State, clock and assumptions

Before calculating, name the state X_t, its units and state space. Then identify the clock: a branching generation is not the same as elapsed time, and a queue's arrival count is not the same as its customer count. Finally specify which increments, individuals or waiting times are independent. These choices determine which theorems apply.

Two common properties concern shifts of time. A process has *stationary increments* if $X_{t+h} - X_t$ has a distribution depending on h but not t. It has *independent increments* if increments over disjoint intervals are independent. Neither property says that the levels X_t are identically distributed.

1.3 A small toolkit

For independent integrable variables, expectation adds: $\mathbb{E}(X+Y) = \mathbb{E}X + \mathbb{E}Y$, whether or not they are independent; independence is needed for $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$. Conditional expectation separates sources of variation:

$$\mathbb{E}X = \mathbb{E}\{\mathbb{E}(X \mid Y)\}, \qquad \operatorname{Var}(X) = \mathbb{E}\{\operatorname{Var}(X \mid Y)\} + \operatorname{Var}\{\mathbb{E}(X \mid Y)\}.$$

We use these identities for branching populations and random walks. For nonnegative X, Markov's inequality follows from $X \geq a\mathbf{1}_{\{X \geq a\}}$:

$$\Pr(X \geq a) \leq \frac{\mathbb{E}X}{a}, \qquad a > 0.$$

For a random variable with finite variance, Chebyshev's inequality applies Markov's inequality to $(X - \mathbb{E}X)^2$:

$$\Pr(|X - \mathbb{E}X| \geq \varepsilon) \leq \frac{\operatorname{Var}(X)}{\varepsilon^2}.$$

These are bounds, not usually exact probabilities. They become useful when an expectation or variance is easy to compute but the full distribution is difficult.

1.4 A modelling checklist

Consider a help desk. The variable $N(t)$ might count calls received by time t, while $Q(t)$ might count calls waiting or being served at time t. The first is nondecreasing; the second rises and falls. Both can be described by stochastic processes, but a Poisson model for N does not make Q Poisson. To model Q one also needs service durations and a rule for what happens when calls overlap.

The state may be a scalar or a vector. If service times are exponential and there is one server, the current count can be a Markov state. If service times are fixed at ten minutes, knowing that one call is in service does not reveal whether it will depart in one second or ten minutes. An augmented state containing elapsed or remaining service time may be needed. The Markov property is therefore a statement about a state representation, not a blanket description of a physical system.

Stationary and independent increments are also distinct. For a random walk with iid increments, both hold. For a nonhomogeneous Poisson process, disjoint increments are independent but not stationary because the event rate changes with calendar time. For a process $X_n = B$ with one random bit repeated at all times, the increments are identically zero after the first

observation, but the levels are strongly dependent. Naming the property precisely prevents the wrong theorem from being invoked.

Worked example 1.3. Suppose independent daily increments Y_i equal 1 with probability .6 and -1 with probability .4, and $S_0 = 0$. Then $\mathbb{E}Y_i = .2$ and $\mathrm{Var}(Y_i) = 1 - .2^2 = .96$, so $\mathbb{E}S_{100} = 20$ and $\mathrm{sd}(S_{100}) = \sqrt{96} \approx 9.80$. A normal approximation addresses S_{100}, a fixed-time quantity. It does not directly answer whether S_n ever fell below -5 during those 100 days.

1.5 What a model can identify

A full model specifies a joint law for all observations of interest. Even when its calculations are exact, the data might not identify its assumptions. For example, a short sequence of arrival counts may fit both a constant-rate Poisson process and a slowly varying-rate process. The models make different predictions about clustering and counts across periods, so those features are more discriminating than a single overall average. This distinction between mathematical consequence and empirical adequacy recurs throughout the book.

Exercises

Exercise 1.1. Construct two processes with the same Bernoulli(p) marginal at every time but different joint distributions at times 1 and 2. Calculate $\Pr(X_1 = X_2)$ for each.

Exercise 1.2. A process counts arrivals during $(0, t]$. Explain why stationary increments does not imply that $N(t)$ has a distribution independent of t.

Chapter 2

Galton–Watson branching

Imagine that each individual independently produces a random number of children, all according to the same offspring distribution. The next generation is the sum of those offspring counts. This simple local rule yields a sharp global question: does the lineage eventually die out?

2.1 Definition and the probability generating function

Let $Z_0 = 1$. For $n \geq 0$, write

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n,i},$$

where the offspring variables $\xi_{n,i}$ are independent and identically distributed nonnegative integers with $\Pr(\xi = k) = p_k$, mean m and variance σ^2. The empty sum is zero, so extinction is absorbing.

The probability generating function (pgf) $f(s) = \mathbb{E}(s^{\xi}) = \sum_{k \geq 0} p_k s^k$ for $0 \leq s \leq 1$ packages the offspring law. It satisfies $f(1) = 1$, $f'(1-) = m$ when the mean is finite, and $f''(1-) = \mathbb{E}[\xi(\xi-1)]$ when this expectation exists. Conditioning on Z_n gives

$$\mathbb{E}(s^{Z_{n+1}} \mid Z_n) = f(s)^{Z_n}.$$

Thus the pgf $f_n(s) = \mathbb{E}(s^{Z_n})$ is the n-fold composition $f_n = f \circ f_{n-1}$ with $f_0(s) = s$. Composition, rather than multiplication, reflects generations nested inside generations.

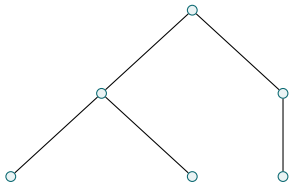


Figure 2.1: One realization of a branching family tree. Each vertex generates its own random offspring count.

2.2 Mean and variance by generation

Conditional on Z_n, the next generation has mean mZ_n and variance $\sigma^2 Z_n$. Consequently

$$\mathbb{E}Z_n = m^n, \qquad v_{n+1} = m^2 v_n + \sigma^2 m^n, \quad v_0 = 0.$$

Solving the recurrence yields, for $m \neq 1$,

$$\operatorname{Var}(Z_n) = \sigma^2 m^{n-1} \frac{m^n - 1}{m - 1},$$

and for $m = 1$, $\operatorname{Var}(Z_n) = n\sigma^2$. The formula is nonnegative also for $m < 1$ because its numerator and denominator have the same sign.

2.3 Extinction as a fixed point

Set $q_n = \Pr(Z_n = 0) = f_n(0)$. Since zero is absorbing, q_n increases and has a limit q, the probability of eventual extinction. Continuity of the pgf on $[0, 1]$ implies $q = f(q)$; iteration from zero shows that q is the *smallest* fixed point in $[0, 1]$.

Theorem 2.1. *For a nondegenerate Galton–Watson process with $p_0 > 0$, if $m \leq 1$ then $q = 1$, while if $m > 1$ then $q < 1$. If $p_0 = 0$, extinction is impossible and $q = 0$. The deterministic case $\xi \equiv 1$ has $m = 1$ but $q = 0$.*

Proof. The pgf is increasing and convex, with $f(1) = 1$ and left derivative m at 1. If $m > 1$ and $p_0 > 0$, then $f(s) - s$ is positive at zero and negative just left of 1, so a smaller fixed point exists. When $m \leq 1$ and the law is nondegenerate, convexity gives $f(s) > s$ for $s < 1$ (except special offspring laws with no chance of extinction, excluded by $p_0 > 0$); hence no fixed point below 1. □

Watch the assumption

The mean controls the extinction threshold, not the extinction probability by itself. Two supercritical offspring laws with the same m can have very different q.

Worked example 2.2. Suppose $\Pr(\xi = 0) = 1/4$ and $\Pr(\xi = 2) = 3/4$. Then $m = 3/2$ and $f(s) = 1/4 + 3s^2/4$. Solving $s = f(s)$ gives $s = 1$ or $s = 1/3$, so $q = 1/3$. Survival has probability $2/3$, even though the expected population grows as $(3/2)^n$.

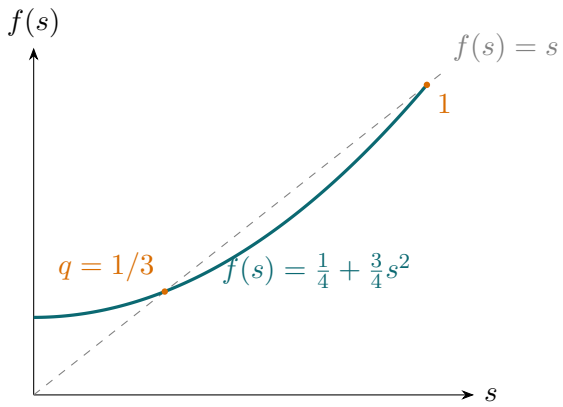


Figure 2.2: Extinction is the smallest intersection of the offspring pgf with the diagonal.

2.4 Subcritical decay and limitations

If $m < 1$, then $\Pr(Z_n > 0) \leq \mathbb{E}Z_n = m^n$ by Markov's inequality. This proves extinction with probability one and gives a simple upper bound on survival to generation n. At criticality the bound is uninformative; the pgf geometry establishes eventual extinction under the stated non-degeneracy assumption. The model assumes independent, identically distributed reproduction and no population-level resource constraint. Density dependence changes the long-run result.

2.5 Other offspring laws

If $\xi \sim \text{Poisson}(\lambda)$, its pgf is $f(s) = e^{\lambda(s-1)}$. The mean is λ and the extinction probability is the smallest solution of $q = e^{\lambda(q-1)}$. There is no elementary closed form in general, but the fixed-point iteration $q_{n+1} = f(q_n)$, starting at $q_0 = 0$, has a direct interpretation: q_n is the probability of extinction by generation n. For $\lambda \leq 1$, the limit is one; for $\lambda > 1$, the limit lies below one.

For a geometric offspring law with $\Pr(\xi = k) = (1-r)r^k$, $k \geq 0$ and $0 < r < 1$, the pgf is $f(s) = (1-r)/(1-rs)$ and $m = r/(1-r)$. Solving $q = f(q)$ gives fixed points 1 and $(1-r)/r$. If $r > 1/2$, the second point is below 1 and is the extinction probability. If $r \leq 1/2$, it is at least 1 and the admissible extinction probability is 1. The threshold is the same $m = 1$ condition as before, but the numerical q comes from the whole pgf.

2.6 Total family size

Let $T = \sum_{n \geq 0} Z_n$, including the initial ancestor. If $m < 1$, monotone convergence allows expectation to pass through the increasing partial sums:

$$\mathbb{E}T = \sum_{n \geq 0} \mathbb{E}Z_n = \sum_{n \geq 0} m^n = \frac{1}{1-m}.$$

At $m \geq 1$, this sum diverges. Infinite expected total size does not imply that an infinite family tree occurs with positive probability: in the critical nondegenerate case extinction is certain, but the total family size has infinite mean. That is a useful warning against treating an infinite expectation as a typical outcome.

The same result follows from a first-generation decomposition. If $T_1, T_2, \ldots$ are independent total descendant counts from each child, then $T \stackrel{d}{=} 1 + \sum_{i=1}^{\xi} T_i$. Taking expectations, when finite, gives $\mathbb{E}T = 1 + m\mathbb{E}T$. This equation has the solution $1/(1-m)$ only when $m < 1$. Applying it naively when $m \geq 1$ would manipulate an infinite expectation as if it were finite.

2.7 Generations and dependence

The family size Z_n is a Markov chain because the next generation depends on the past only through the number of current individuals, under the iid reproduction assumption. It is generally *not* a random walk: the distribution of $Z_{n+1} - Z_n$ depends on Z_n, since more current individuals produce more offspring opportunities. The conditional mean drift is $(m-1)Z_n$, growing with population size. A density-dependent biological process would change this drift again, perhaps making a large population less productive.

Exercises

Exercise 2.1. For offspring law $\Pr(\xi = 0) = 2/5$, $\Pr(\xi = 1) = 1/5$, $\Pr(\xi = 2) = 2/5$, find m, $f(s)$ and the extinction probability. Explain the result.

Exercise 2.2. Starting from $Z_0 = r$ independent ancestors, express the extinction probability in terms of the one-ancestor probability q.

Exercise 2.3. Derive the variance recurrence using the law of total variance, and check its critical-case limit as $m \to 1$.

Chapter 3

Random walks and first passage

A random walk accumulates random increments. In a gambling interpretation, its state is current capital, and the interesting question is often not its value at a fixed time but which boundary it reaches first.

3.1 Accumulation and scale

Let $Y_1, Y_2, \dots$ be iid increments and $S_n = x + \sum_{i=1}^{n} Y_i$. Then $\mathbb{E}S_n = x + n\mu$ and $\mathrm{Var}(S_n) = n\sigma^2$ when $\mu = \mathbb{E}Y_1$ and $\sigma^2 = \mathrm{Var}(Y_1)$ are finite. The law of large numbers gives $(S_n - x)/n \to \mu$ almost surely under standard integrability conditions. The central limit theorem gives

$$\frac{S_n - x - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1)$$

when $0 < \sigma^2 < \infty$. Thus drift grows as n and typical fluctuations as $\sqrt{n}$. Neither statement directly determines a first-passage probability, which concerns an entire path.

3.2 Gambler's ruin

Take $Y_i = +1$ with probability p and -1 with probability $q = 1 - p$. Fix boundaries 0 and d and an initial state $x \in \{0, \dots, d\}$. Let $\tau = \inf\{n \geq 0 : S_n \in \{0, d\}\}$ and $h_x = \Pr_x(S_\tau = d)$. Conditioning on the first step,

$$h_x = p h_{x+1} + q h_{x-1}, \quad 1 \leq x < d; \qquad h_0 = 0, \quad h_d = 1.$$

For $p = q = 1/2$, the second difference is zero, so h_x is linear and

$$\boxed{h_x = x/d.}$$

For $p \neq q$, the characteristic roots are 1 and q/p, giving

$$h_x = \frac{1 - (q/p)^x}{1 - (q/p)^d}.$$

The symmetric expression is the continuous limit as $p \to q$.

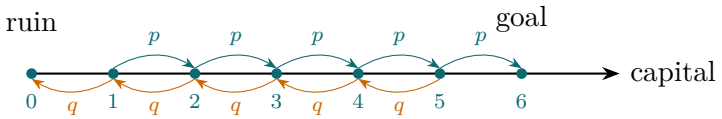


Figure 3.1: First-step analysis converts a path question into a boundary-value recurrence.

3.3 Expected duration

Write $e_x = \mathbb{E}_x\tau$. The first-step equation is $e_x = 1 + pe_{x+1} + qe_{x-1}$ with $e_0 = e_d = 0$. In the symmetric case, a quadratic trial function works:

$$\boxed{e_x = x(d - x).}$$

The expected duration is largest near the middle. For $p \neq q$, a particular linear solution and the homogeneous solutions give

$$e_x = \frac{x}{q - p} - \frac{d}{q - p}\,\frac{1 - (q/p)^x}{1 - (q/p)^d}.$$

This expression has a removable singularity at $p = q$; use $x(d - x)$ there.

Worked example 3.1. With fair steps, $d = 10$ and $x = 3$, the chance of reaching 10 first is $3/10$, while the expected number of bets before absorption is $3(10 - 3) = 21$. A probability and an expected time answer different questions.

Watch the assumption

The finite-interval formulas rely on absorption at either boundary and steps of exactly one unit. On the unbounded integer line, recurrence, return times and drift require separate analysis.

3.4 Solving the recurrences from scratch

The first-step equation can be rearranged as

$$p(h_{x+1} - h_x) = q(h_x - h_{x-1}).$$

Thus the successive differences $\Delta_x = h_x - h_{x-1}$ form a geometric sequence with ratio q/p. Summing the differences from 1 to x and using $h_0 = 0$ gives

$$h_x = \Delta_1 \sum_{k=0}^{x-1} (q/p)^k.$$

The condition $h_d = 1$ determines Δ_1. This derivation makes the symmetric case transparent: when $p = q$, every difference is the same, so h_x is linear.

For expected duration, rearrange the fair recurrence as $e_{x+1} - 2e_x + e_{x-1} = -2$. A quadratic has constant second difference, which suggests $e_x = Ax^2 + Bx + C$. The boundary values give $C = 0$ and $B = -Ad$, while the second difference gives $2A = -2$. Therefore $e_x = x(d - x)$. This is a standard problem-solving pattern: the form of the difference equation suggests a trial solution, then boundaries determine constants.

3.5 Why the boundary problem is well posed

There are only $d-1$ interior states. Starting from any one of them, a run of at most d consecutive upward steps reaches d, and a run of at most d downward steps reaches 0. When $0 < p < 1$, either run has a strictly positive probability uniformly over the current interior state. Repeating blocks of d steps bounds the tail of τ by a geometric sequence. Hence absorption occurs almost surely and $\mathbb{E}\tau < \infty$. This justifies the expected-duration equation and its finite solution.

At a biased boundary, the expression for e_x should be checked by limits and signs. If $p > q$, the walk tends upward, and h_x should exceed x/d; if $p < q$, it should be below. The formula does both. At $x = 0$ it gives zero, and at $x = d$ it gives one. Such basic checks catch algebraic slips faster than repeating the derivation.

Exercises

Exercise 3.1. For a fair walk between 0 and 8 starting at 5, compute the chance of reaching 8 before 0 and the expected absorption time.

Exercise 3.2. For $p = 2/3$, $d = 4$ and $x = 1$, calculate the goal-hitting probability. Check whether it is above or below the fair-walk value.

Exercise 3.3. Show directly that $x(d-x)$ satisfies the symmetric expected-duration recurrence and both boundary conditions.

Chapter 4

Discrete-time Markov chains

The Markov property says that a well-chosen current state contains the information needed to predict the next state. It does not say that successive states are independent. For a queue, the current number waiting may be sufficient under memoryless arrival and service assumptions; under other service models, elapsed service time may also be needed.

4.1 Transition rules

Definition 4.1. A discrete-time Markov chain on a finite or countable state space S satisfies

$$\Pr(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = \Pr(X_{n+1} = j \mid X_n = i) = p_{ij}.$$

For a time-homogeneous chain, p_{ij} does not depend on n. Its transition matrix $\boldsymbol{P} = (p_{ij})$ has nonnegative entries and every row sums to one.

If the row vector $\boldsymbol{\mu}_0$ gives the initial law, then $\boldsymbol{\mu}_n = \boldsymbol{\mu}_0 \boldsymbol{P}^n$. The n-step transition probability is $p_{ij}^{(n)} = (\boldsymbol{P}^n)_{ij}$. Conditioning on the intermediate state gives the Chapman–Kolmogorov equations:

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}.$$

Matrix multiplication is therefore a probabilistic sum over all possible intermediate states.

Worked example 4.2. For a two-state weather model, let

$$\boldsymbol{P} = \begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix},$$

with states sunny and rainy. Starting sunny, the probability of rain after two days is $(\boldsymbol{P}^2)_{12} = .8(.2) + .2(.7) = .30$. The two terms correspond to the two possible states after one day.

4.2 Hitting questions use the same principle

For target set $A \subseteq S$, let $T_A = \inf\{n \geq 0 : X_n \in A\}$. If $h_i = \Pr_i(T_A < T_B)$ for disjoint sets A, B, then

$$h_i = \sum_j p_{ij} h_j \quad (i \notin A \cup B), \qquad h_i = 1 \ (i \in A), \quad h_i = 0 \ (i \in B).$$

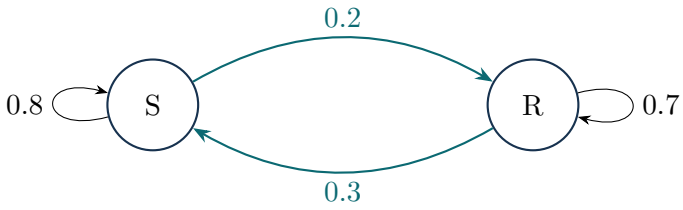


Figure 4.1: A transition diagram for the two-state weather chain. Outgoing probabilities from each state sum to one.

Similarly, if the mean hitting time is finite, $e_i = \mathbb{E}_i T_A$ solves

$$e_i = 1 + \sum_j p_{ij} e_j \quad (i \notin A), \qquad e_i = 0 \quad (i \in A).$$

These are the Markov-chain versions of the random-walk equations in Chapter 3.

4.3 Stationary distributions

Definition 4.3. A probability row vector $\boldsymbol{\pi}$ is stationary if $\boldsymbol{\pi}\boldsymbol{P} = \boldsymbol{\pi}$. Starting the chain with $X_0 \sim \boldsymbol{\pi}$ then gives $X_n \sim \boldsymbol{\pi}$ at every time.

For the weather chain, writing $\pi_S + \pi_R = 1$ and equating flow gives $.2\pi_S = .3\pi_R$, so $\boldsymbol{\pi} = (.6, .4)$. The initial state affects short-term probabilities but not the stationary equation.

If $\boldsymbol{P}$ is doubly stochastic, its columns as well as rows sum to one, and the uniform distribution is stationary on a finite state space. This condition is sufficient, not necessary.

Why it works

Multiplying by $\boldsymbol{P}$ moves a probability distribution forward one step. A stationary distribution is a fixed point of that map. Whether an arbitrary starting distribution approaches that fixed point is a separate question.

4.4 Spectral view for two states

For $\boldsymbol{P} = \left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right)$ with $a, b > 0$, the eigenvalues are 1 and $1 - a - b$. The stationary law is $(b/(a+b), a/(a+b))$. The second eigenvalue controls how deviations decay: when $|1-a-b| < 1$, $\boldsymbol{P}^n$ approaches a matrix with every row equal to $\boldsymbol{\pi}$. At $a = b = 1$ it equals -1 and the chain alternates forever.

4.5 Reading a transition matrix

Rows correspond to the current state and columns to the next state. This orientation matters when using row probability vectors. For the weather model, the two-step sunny-to-rainy probability is .30, but the two-step rainy-to-sunny probability is $(\boldsymbol{P}^2)_{21} = .3(.8) + .7(.3) = .45$. The entries need not be symmetric, and a row-stochastic matrix need not be column-stochastic.

A transition diagram can make impossible moves visible. For example, a three-state chain with matrix

$$\boldsymbol{P} = \begin{pmatrix} .7 & .3 & 0 \\ .2 & .5 & .3 \\ 0 & .4 & .6 \end{pmatrix}$$

can move only between neighbours or remain where it is. A two-step transition from state 1 to state 3 is possible through state 2, with probability $.3(.3) = .09$, even though the one-step probability is zero. Reachability is a property of powers of $\boldsymbol{P}$, not only of its immediate nonzero entries.

4.6 The Markov assumption as a testable simplification

Suppose that tomorrow's weather depends on today's weather and also on whether it has rained for three consecutive days. A two-state description cannot express this memory, but a state such as (current weather, current run length) can. The state space becomes larger while the transition rule becomes local in time. This illustrates a general modelling tradeoff: compression of the past into a state makes prediction tractable, but too much compression discards relevant information.

Time homogeneity is another separate assumption. A chain may be Markov with morning and evening transition matrices $\boldsymbol{P}_n$ that vary with n. Then the law evolves by $\boldsymbol{\mu}_n = \boldsymbol{\mu}_0 \boldsymbol{P}_0 \boldsymbol{P}_1 \cdots \boldsymbol{P}_{n-1}$, not by a single matrix power. Solving $\boldsymbol{\pi}\boldsymbol{P} = \boldsymbol{\pi}$ for one matrix would not describe such a process.

4.7 Stationarity versus equilibrium from one start

Suppose the initial distribution equals $\boldsymbol{\pi}$. Then stationarity means every later *marginal* is $\boldsymbol{\pi}$, even though the path still changes. It does not mean the chain freezes. Conversely, if the chain starts in state i, its law may take time to approach $\boldsymbol{\pi}$. In a periodic chain, it may never approach at individual times despite the existence of a stationary law. The next chapter identifies the conditions.

Exercises

Exercise 4.1. For $\boldsymbol{P} = \left(\begin{smallmatrix} .9 & .1 \\ .4 & .6 \end{smallmatrix}\right)$, calculate $(\boldsymbol{P}^2)_{12}$ and the stationary distribution.

Exercise 4.2. A three-state chain moves from 1 to 2, 2 to 3 and 3 to 1 deterministically. Find a stationary distribution and describe $\Pr(X_n = 1 \mid X_0 = 1)$ as n grows.

Exercise 4.3. Prove directly that the uniform distribution is stationary for a finite doubly stochastic matrix.

Chapter 5

Classes, recurrence and reversibility

Long-run claims require more than solving $\boldsymbol{\pi P} = \boldsymbol{\pi}$. A chain can have several closed regions, or it can cycle with a fixed rhythm. Class structure and period explain which conclusions are available.

5.1 Communication and closed classes

State j is accessible from i, written $i \to j$, if $p_{ij}^{(n)} > 0$ for some $n \geq 0$. States communicate if each is accessible from the other. Communication is an equivalence relation, so it partitions S into classes. A class is *closed* if no transition from it reaches outside it. A chain with one communicating class is *irreducible*. An absorbing state is a one-state closed class with $p_{ii} = 1$.

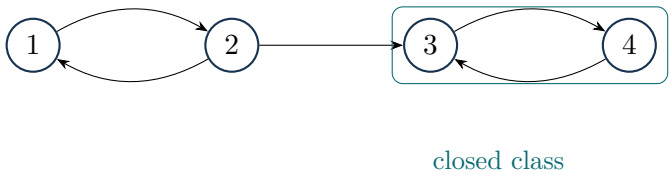


Figure 5.1: The chain can leave $\{1, 2\}$ but cannot leave $\{3, 4\}$.

5.2 Return and period

Let $T_i^+ = \inf\{n \geq 1 : X_n = i\}$. State i is recurrent if $\Pr_i(T_i^+ < \infty) = 1$ and transient otherwise. A recurrent state is positive recurrent if $\mathbb{E}_i T_i^+ < \infty$, otherwise null recurrent. In a finite chain, every state in a closed irreducible class is positive recurrent; states outside all closed classes are transient.

The period of i is $d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}$. Communicating states share a period. Period one means aperiodic. A self-loop is sufficient for aperiodicity in an irreducible class, but it is not necessary for every individual state to have a self-loop.

Theorem 5.1. *A finite irreducible chain has a unique stationary distribution with strictly positive entries. If it is also aperiodic, then $p_{ij}^{(n)} \to \pi_j$ for every i, j.*

The aperiodicity assumption belongs to the *limit*, not to existence of $\boldsymbol{\pi}$. For deterministic alternation between two states, $\boldsymbol{\pi} = (1/2, 1/2)$ is stationary, but $\boldsymbol{P}^n$ oscillates.

5.3 Global and detailed balance

The stationarity equation at state j,

$$\pi_j = \sum_i \pi_i p_{ij},$$

equates total probability mass arriving at j with the mass currently there. Detailed balance imposes the stronger pairwise equations

$$\pi_i p_{ij} = \pi_j p_{ji} \qquad \text{for all } i, j.$$

Summing over i shows that detailed balance implies stationarity. A chain satisfying detailed balance is reversible when started in its stationary law: a stationary path has the same distribution when its time order is reversed. Stationarity alone does not imply reversibility.

If $\pi_i > 0$, the reversed transition probabilities are $p^*_{ij} = \pi_j p_{ji}/\pi_i$. Detailed balance says $\boldsymbol{P}^* = \boldsymbol{P}$. For a finite irreducible chain, Kolmogorov's cycle criterion provides another test: the product of transition probabilities around every directed cycle equals the product around the reversed cycle, counting a missing reverse transition as probability zero.

5.4 Two useful families

For a simple random walk on an undirected graph, $p_{ij} = 1/\deg(i)$ for neighbours. Therefore $\pi_i = \deg(i)/(2|E|)$ satisfies detailed balance because $\pi_i p_{ij} = 1/(2|E|)$ on every edge. The degree, not uniformity, determines long-run occupancy unless the graph is regular.

For a nearest-neighbour birth–death chain with upward probability p_i and downward probability q_i, local balance gives $\pi_i p_i = \pi_{i+1} q_{i+1}$. Hence

$$\pi_n = \pi_0 \prod_{k=0}^{n-1} \frac{p_k}{q_{k+1}},$$

provided the resulting positive weights have a finite normalising sum. A formal solution with infinite total mass is not a stationary probability distribution.

5.5 Why a stationary law can fail to be unique

Consider two absorbing states, 0 and 1, with no transitions between them. Both point masses δ_0 and δ_1 are stationary, and so is every mixture $a\delta_0 + (1-a)\delta_1$. The long-run law from a given initial state depends on which closed class the path enters. A unique stationary law for the *whole* finite chain therefore cannot be asserted unless the closed-class structure supports it.

Even when several classes exist, each finite closed irreducible class has its own stationary law. Any stationary law on the whole chain is a convex combination of those class laws, with zero mass on transient states. This gives a structural way to solve stationarity before attempting linear algebra. It also explains why a transition matrix may have more than one linearly independent left eigenvector for eigenvalue 1.

5.6 Period and time averages

A deterministic three-cycle has period 3: starting at state 1, its distribution cycles among point masses. Yet it spends exactly one third of its time in each state over any complete set of cycles.

For a finite irreducible chain, long-run empirical visit frequencies converge to the stationary probabilities even if the one-time transition probabilities oscillate. Thus two different limits are in play: a time average along a path and a distribution at a distant single time.

The return-time identity gives another interpretation. In a finite irreducible chain, $\pi_i = 1/\mathbb{E}_i T_i^+$. A state visited often has a short mean recurrence time. This is consistent with the graph-walk result: high-degree vertices receive more stationary mass and have shorter mean return times. The identity is useful for checking a solved stationary distribution against a return-time calculation.

5.7 Testing reversibility with a cycle

For a directed three-cycle that moves $1 \to 2 \to 3 \to 1$ deterministically, the uniform law is stationary. It cannot be reversible: the forward product around the cycle is 1, while the reverse product is 0. More generally, detailed balance asks whether every pairwise flow can be balanced. A chain may satisfy total balance at every vertex while retaining a net circulating current around a cycle.

On a tree, by contrast, there is no nontrivial cycle to support a circulating current. For an irreducible nearest-neighbour walk on a finite undirected tree, one can assign stationary weights recursively along edges and normalise them, obtaining reversibility. The absence of cycles is a structural reason local balance is especially natural for birth–death chains, whose graph is a path.

Exercises

Exercise 5.1. Find the communication classes, closed classes and stationary distributions of the chain $1 \to 2$, $2 \to 1$ with probability $1/2$ and $2 \to 3$ with probability $1/2$, and $3 \to 3$.

Exercise 5.2. For a random walk on a path with four vertices, find the stationary distribution and identify the period. Does $\boldsymbol{P}^n$ converge?

Exercise 5.3. Use detailed balance to find the stationary distribution of a three-state birth–death chain with $p_0 = 1$, $q_1 = 1/3$, $p_1 = 2/3$ and $q_2 = 1$.

Part II

Events and transitions in continuous time

Chapter 6

Poisson processes in time and space

A Poisson process models events occurring at a constant average rate when counts on disjoint time intervals are independent. It is a model for *events*, not automatically for the number of customers present in a queue. That distinction will matter in Chapter 8.

6.1 Homogeneous counts and waiting times

Definition 6.1. A homogeneous Poisson process $N(t)$ with rate $\lambda > 0$ has $N(0) = 0$, independent stationary increments, and

$$N(t+h) - N(t) \sim \text{Poisson}(\lambda h), \qquad t, h \geq 0.$$

Consequently $\mathbb{E}N(t) = \text{Var}(N(t)) = \lambda t$ and

$$\Pr\{N(t) = n\} = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

Let T_1 be the first arrival time. The event $T_1 > t$ is exactly $N(t) = 0$, so

$$\Pr(T_1 > t) = e^{-\lambda t}, \quad T_1 \sim \text{Exp}(\lambda).$$

Independent increments yield independent exponential interarrival times $W_1, W_2, \ldots$, each with mean $1/\lambda$. Their sums $T_n = \sum_{i=1}^{n} W_i$ are arrival times. Conversely, iid exponential interarrivals construct a Poisson process. The exponential memoryless identity

$$\Pr(W > s + t \mid W > s) = \Pr(W > t)$$

is why the elapsed wait need not be part of the state in the basic Markov models.

6.2 Forward equation and a rare-event limit

Let $p_n(t) = \Pr(N(t) = n)$. In a short interval h, the probability of one event is $\lambda h + o(h)$ and of two or more events is $o(h)$. Thus

$$p_0'(t) = -\lambda p_0(t), \qquad p_n'(t) = \lambda p_{n-1}(t) - \lambda p_n(t), \quad n \geq 1,$$

with $p_0(0) = 1$, $p_n(0) = 0$ for $n \geq 1$. The Poisson probabilities solve these equations. A related construction divides time t into m slots, each with event probability $\lambda t/m$. The binomial count converges to Poisson(λt) as $m \to \infty$. The limit motivates the model but does not verify its assumptions in a real event stream.

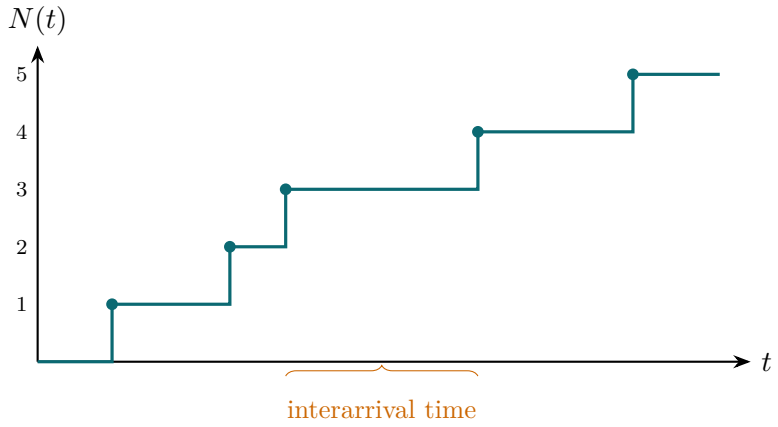


Figure 6.1: A Poisson count is a right-continuous step process with unit jumps.

6.3 Conditional arrival times

Conditional on $N(t) = n$, the unordered event times are iid Uniform$(0, t)$, and the ordered times are their order statistics. For example, given exactly two arrivals in $(0, t]$, the probability both occur before $t/2$ is $(1/2)^2 = 1/4$. Unconditionally the two arrival times are not uniform on a fixed interval.

6.4 Changing rates and transforming time

For a nonhomogeneous Poisson process with deterministic locally integrable rate $\lambda(u) \geq 0$, define the cumulative intensity $\Lambda(t) = \int_0^t \lambda(u)\, du$. Independent increments remain, but their means are $\Lambda(t) - \Lambda(s)$ and stationarity of increments generally fails:

$$N(t) - N(s) \sim \text{Poisson}\left(\int_s^t \lambda(u)\, du \right).$$

The waiting time after time s has survival probability $\exp[-\int_s^{s+w} \lambda(u)\, du]$. It is generally not exponential in w; successive calendar-time waits need not be identically distributed. One can simulate by transforming a unit-rate Poisson process through the inverse of Λ, when an inverse is available, or by thinning a dominating homogeneous process.

6.5 Superposition and thinning

Independent Poisson processes with rates λ_1, λ_2 superpose to a Poisson process with rate $\lambda_1 + \lambda_2$. Independently retaining each event with probability r produces a Poisson process of rate $r\lambda$; rejected and retained processes are independent. If retention varies with time, the retained process has intensity $r(t)\lambda(t)$.

6.6 Spatial Poisson processes

On a plane, a homogeneous spatial Poisson process of intensity λ assigns each bounded region A a Poisson$(\lambda|A|)$ count, with independent counts for disjoint regions. Here λ has units of

points per area. If R is the distance from a fixed location to the nearest point, then $\{R > r\}$ means the radius-r disk is empty:

$$\Pr(R > r) = e^{-\lambda \pi r^2}, \qquad F_R(r) = 1 - e^{-\lambda \pi r^2} \quad (r \geq 0).$$

For varying spatial intensity $\lambda(x)$, replace $\lambda|A|$ with $\int_A \lambda(x)\, dx$. The void-probability argument still works.

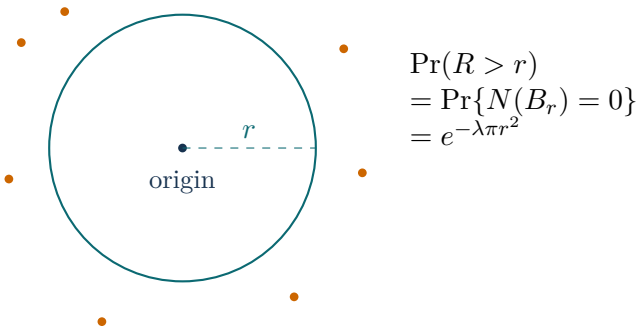


Figure 6.2: The nearest point lies beyond radius r exactly when the disk B_r is empty.

6.7 Counting and waiting are two views of one model

The event $\{N(t) \geq n\}$ is exactly $\{T_n \leq t\}$, where T_n is the nth arrival time. Hence

$$\Pr(T_n \leq t) = 1 - \sum_{k=0}^{n-1} e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

Since T_n is a sum of n independent $\text{Exp}(\lambda)$ waits, it also has a Gamma distribution with shape n and rate λ. The count and wait formulas are two descriptions of the same sample path; they provide a useful cross-check. For instance, the chance that the third event occurs by time t equals the chance of at least three counts by t.

6.8 A nonhomogeneous worked example

Suppose the event rate is $\lambda(t) = 2 + t$ per hour for $t \geq 0$. The expected number of events in the first two hours is

$$\Lambda(2) = \int_0^2 (2 + t)\, dt = 6.$$

The count $N(2)$ is Poisson(6), and $\Pr(N(2) = 0) = e^{-6}$. From hour 1 to hour 2, the expected count is $\Lambda(2) - \Lambda(1) = 6 - 2.5 = 3.5$. By contrast, a homogeneous model using the initial rate 2 would predict only 2 events in that hour. Choosing a rate from the wrong time interval gives a mathematically correct calculation for the wrong model.

To simulate the first arrival, draw $E \sim \text{Exp}(1)$ and solve $\Lambda(T_1) = E$. Here $\Lambda(t) = 2t + t^2/2$, so $T_1 = -2 + \sqrt{4 + 2E}$. For later arrivals, add independent unit-rate exponential amounts on the cumulative-intensity scale and invert again. This construction shows why the transformed time $\Lambda(t)$ is useful.

6.9 Thinning and marked events

Suppose calls arrive at rate λ and each is independently a technical-support call with probability r. The technical calls form a Poisson process of rate $r\lambda$, and the remaining calls form an independent Poisson process of rate $(1-r)\lambda$. The independence conclusion relies on independent marking of each event. If a system-wide outage makes many calls technical at once, the marks are dependent and ordinary thinning no longer describes the data.

In space, the same idea classifies points. If a planar Poisson pattern of intensity λ is independently retained with location-dependent probability $r(x)$, the retained intensity is $\lambda r(x)$. The count in a region A has mean $\int_A \lambda r(x)\,dx$. This can model detection probability varying across a survey area, but spatial clustering in the original process would remain a separate model issue.

Exercises

Exercise 6.1. For rate $\lambda = 3$ per hour, compute the probability of exactly two events in 30 minutes and the probability the first wait exceeds 20 minutes.

Exercise 6.2. Given $N(2) = 3$, find the chance that all three events occurred in the first hour. State the conditioning argument.

Exercise 6.3. For a planar intensity of 2 points per square kilometre, find the median distance from a fixed location to its nearest point.

Chapter 7

Continuous-time Markov chains

In a continuous-time Markov chain (CTMC), transitions can occur at any time. The state holds for a random duration and then jumps. The generator records transition *rates*; its rows sum to zero, unlike a transition matrix's rows.

7.1 Holding times, jump chain and generator

For a finite-state CTMC, let $q_{ij} \geq 0$ for $j \neq i$ be the rate of moving from i to j and set $q_{ii} = -\sum_{j \neq i} q_{ij}$. The matrix $\boldsymbol{Q} = (q_{ij})$ is the generator. In state i, the holding time is exponential with rate $\nu_i = -q_{ii}$. If $\nu_i > 0$, the next state is j with probability $k_{ij} = q_{ij}/\nu_i$. The matrix $\boldsymbol{K}$ is the embedded jump chain. An absorbing state has $\nu_i = 0$.

This follows from the short-time rules

$$\begin{aligned} \Pr(X(t+h) = j \mid X(t) = i) &= q_{ij}h + o(h) && (j \neq i), \\ \Pr(X(t+h) = i \mid X(t) = i) &= 1 - \nu_i h + o(h). \end{aligned}$$

For finite state spaces the rates are bounded, so the chain is nonexplosive: infinitely many jumps cannot accumulate in finite time. Infinite-state models need a separate regularity check.

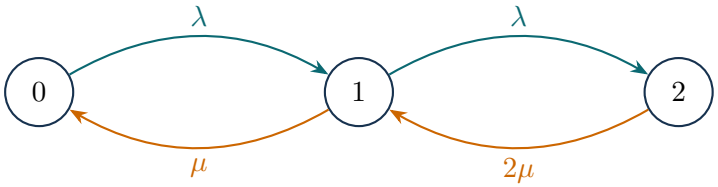


Figure 7.1: A transition-rate diagram labels arrows with rates, not one-step probabilities.

7.2 Evolution equations

Let $\boldsymbol{P}(t) = (p_{ij}(t))$, where $p_{ij}(t) = \Pr(X(t) = j \mid X(0) = i)$. The Chapman–Kolmogorov identity is $\boldsymbol{P}(s+t) = \boldsymbol{P}(s)\boldsymbol{P}(t)$. Differentiating on the right or left gives the Kolmogorov equations

$$\boldsymbol{P}'(t) = \boldsymbol{P}(t)\boldsymbol{Q} \quad \text{(forward)}, \qquad \boldsymbol{P}'(t) = \boldsymbol{Q}\boldsymbol{P}(t) \quad \text{(backward)}, \qquad \boldsymbol{P}(0) = \boldsymbol{I}.$$

For a finite-state homogeneous CTMC, their unique solution is $\boldsymbol{P}(t) = e^{t\boldsymbol{Q}}$. The two products agree because the exponential is a power series in $\boldsymbol{Q}$. The forward equation conditions on the

state just before time t; the backward equation conditions on the first transition after time zero.

7.3 The two-state model

Suppose $0 \to 1$ at rate α and $1 \to 0$ at rate β. Then

$$\boldsymbol{Q} = \begin{pmatrix} -\alpha & \alpha \\ \beta & -\beta \end{pmatrix}, \qquad \boldsymbol{\pi} = \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta} \right).$$

Starting in 0, solve $u'(t) = \alpha - (\alpha+\beta)u(t)$ for $u(t) = \Pr(X(t) = 1)$:

$$u(t) = \frac{\alpha}{\alpha+\beta}\left(1 - e^{-(\alpha+\beta)t}\right).$$

The stationary probability is the long-time limit. This finite irreducible CTMC converges despite its jump chain alternating between the two states: random holding times remove the discrete-time parity obstruction.

7.4 Stationarity, balance and recurrence

A probability row vector is stationary when $\boldsymbol{\pi}\boldsymbol{Q} = \mathbf{0}$. Detailed balance is $\pi_i q_{ij} = \pi_j q_{ji}$ for all $i \neq j$; summing gives stationarity. In a finite irreducible CTMC, the stationary law is unique and $\boldsymbol{P}(t)$ converges to it as $t \to \infty$. For countably infinite chains, positive recurrence and nonexplosion matter; a formal solution to $\boldsymbol{\pi}\boldsymbol{Q} = 0$ must be nonnegative and normalisable.

The Poisson process is a CTMC on $\{0, 1, \ldots\}$ with $q_{n,n+1} = \lambda$, $q_{nn} = -\lambda$. It has no stationary probability distribution because its count drifts upward without return.

7.5 Constructing a sample path

The rate diagram specifies a direct simulation algorithm. At current state i, draw a holding time $H \sim \mathrm{Exp}(\nu_i)$ and independently draw the next state from the row $k_{ij} = q_{ij}/\nu_i$. Advance the clock by H, move to that state, and repeat. The exponential holding time is memoryless, so conditional on the current state, the residual waiting time has the same law regardless of how long the process has already stayed there.

Competing clocks give the same construction. Suppose at state i there are independent potential transitions to j with exponential rates q_{ij}. The minimum of these clocks is exponential with rate $\nu_i = \sum_{j \neq i} q_{ij}$, and clock j wins with probability q_{ij}/ν_i. This explains both the total exit rate and the jump-chain probabilities. It also prevents a common error: q_{ij} is a rate with units inverse time, so it need not lie below one.

7.6 Forward equations state by state

Let $\mu_j(t) = \Pr(X(t) = j)$ for a fixed initial distribution. The forward equation reads

$$\mu_j'(t) = \sum_{i \neq j} \mu_i(t) q_{ij} - \mu_j(t) \sum_{k \neq j} q_{jk}.$$

The first term is probability flow into j; the second is flow out. Summing over j gives zero, as total probability must remain one. For the two-state model, $\mu_1' = \alpha\mu_0 - \beta\mu_1$ and $\mu_0 = 1 - \mu_1$, producing the differential equation solved earlier. This is a continuous-time analogue of the discrete stationary equation, except that it describes evolution at every time.

At stationarity, the derivative is zero and the flow equations become $\boldsymbol{\pi}\boldsymbol{Q} = \mathbf{0}$. These are *global* balance equations. Reversibility adds pairwise balance. A chain may have a stationary circulation on a directed cycle, so the two concepts must remain separate.

7.7 Uniformisation for a finite generator

Choose $\gamma \geq \max_i \nu_i$ and set $\boldsymbol{R} = \boldsymbol{I} + \boldsymbol{Q}/\gamma$. The matrix $\boldsymbol{R}$ is stochastic: its off-diagonal entries are nonnegative, its diagonal entries are $1 - \nu_i/\gamma$, and its rows sum to one. Since $\boldsymbol{Q} = \gamma(\boldsymbol{R} - \boldsymbol{I})$,

$$e^{t\boldsymbol{Q}} = e^{-\gamma t} \sum_{n=0}^{\infty} \frac{(\gamma t)^n}{n!} \boldsymbol{R}^n.$$

Thus a finite CTMC transition law is a Poisson-weighted mixture of discrete-chain steps. Some steps of $\boldsymbol{R}$ may be self-transitions added by the construction. This representation is useful for numerical calculation and shows the link between Poisson clocks and CTMCs; it is not the same as the embedded jump chain $\boldsymbol{K}$.

Exercises

Exercise 7.1. For the two-state CTMC with $\alpha = 2$ and $\beta = 3$, find its stationary law and $\Pr(X(t) = 1 \mid X(0) = 0)$.

Exercise 7.2. Write the generator and jump-chain probabilities for states $\{0, 1, 2\}$ with rates $0 \to 1$ equal to 2, $1 \to 0$ equal to 1, $1 \to 2$ equal to 3, and $2 \to 1$ equal to 4.

Exercise 7.3. Explain why the diagonal of a generator is nonpositive and why each row sums to zero.

Chapter 8

Birth–death chains and queues

Queueing models combine arrival rates with service capacity. The state is usually the number of customers in the system, including those being served. A birth raises that count by one; a death lowers it by one. The key constraint is whether long-run service capacity can keep up with arrivals.

8.1 Birth–death balance

For states $n = 0, 1, \ldots$, let the birth rate at n be λ_n and the death rate at $n \geq 1$ be μ_n. The generator has $q_{n,n+1} = \lambda_n$, $q_{n,n-1} = \mu_n$ and $q_{nn} = -(\lambda_n + \mu_n)$, with $\mu_0 = 0$. Local balance gives

$$\pi_n \lambda_n = \pi_{n+1} \mu_{n+1}, \qquad \pi_n = \pi_0 \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}.$$

To obtain a probability distribution, normalise these weights:

$$\pi_0 = \left[1 + \sum_{n=1}^{\infty} \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}} \right]^{-1},$$

provided the bracketed sum is finite. The normalising condition is a substantive stability condition, not algebraic housekeeping.

8.2 $M/M/1$: one server

Arrivals form a Poisson process with rate λ and service durations are independent $\mathrm{Exp}(\mu)$. With $n > 0$ customers, a single server completes work at rate μ. Thus $\lambda_n = \lambda$, $\mu_n = \mu$ for $n \geq 1$. Setting $\rho = \lambda/\mu$ gives

$$\pi_n = (1 - \rho)\rho^n, \quad n \geq 0, \qquad \text{if and only if } \rho < 1.$$

The mean number in the system is $L = \rho/(1 - \rho)$. If arrivals see the stationary time average (as Poisson arrivals do under the usual independence assumptions), Little's law gives mean time in system $W = L/\lambda = 1/(\mu - \lambda)$. The formula diverges as load approaches capacity.

Why it works

At high utilisation, spare capacity is small. Random arrival and service fluctuations then create long busy spells even when the mean arrival rate is still below the mean service rate.

8.3 $M/M/\infty$: unlimited parallel service

With infinitely many servers, every customer starts service immediately. At state n, n independent services are in progress, so the total departure rate is $n\mu$. Local balance yields

$$\pi_n = e^{-\lambda/\mu}\frac{(\lambda/\mu)^n}{n!}.$$

The stationary number is Poisson with mean λ/μ. There is no finite-server overload threshold because service capacity grows with n.

8.4 $M/M/c$ and finite capacity

With c parallel servers and one common queue, the total departure rate is $\mu_n = \min(n, c)\mu$. The birth rate remains λ. For $n \geq c$, the stationary weights have geometric ratio $\lambda/(c\mu)$; thus a stationary probability distribution exists precisely when $\lambda < c\mu$. An explicit normalisation is

$$\pi_0^{-1} = \sum_{n=0}^{c-1}\frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^c}{c!}\frac{1}{1-\lambda/(c\mu)}.$$

For a finite-capacity $M/M/c/K$ system, states are $0, \ldots, K$ and arrivals finding K customers are blocked. The finite irreducible chain has a stationary distribution for every positive λ, μ; its throughput is $\lambda(1-\pi_K)$, not the offered rate λ.

8.5 A repair model

Two identical machines can fail independently at rate λ while operating, and one repairperson repairs at rate μ. If n is the number of failed machines, the states are $0, 1, 2$. Birth rates are $\lambda_0 = 2\lambda$, $\lambda_1 = \lambda$; death rates are $\mu_1 = \mu_2 = \mu$. Local balance gives unnormalised weights $1, 2\lambda/\mu, 2(\lambda/\mu)^2$. The finite state space always permits normalisation. Notice why the failure rate decreases as machines fail: only operating machines can fail.

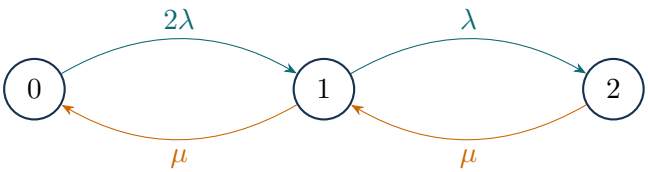


Figure 8.1: Number of failed machines when one repairperson serves both machines.

8.6 Comparing capacity models

The three basic queue families have the same arrival rate λ but different service capacity functions:

model	total departure rate at state n
$M/M/1$	$\mu\,\mathbf{1}_{\{n>0\}}$
$M/M/c$	$\min(n, c)\mu$
$M/M/\infty$	$n\mu$.

This table is often more informative than memorising stationary formulas. In $M/M/1$, once the server is busy, adding customers does not raise service capacity. In $M/M/c$, capacity rises until all c servers are busy and then saturates. In $M/M/\infty$, it keeps rising. The stability conditions follow from the tail of the birth–death normalising sum.

For example, take $\lambda = 4$ and $\mu = 3$ per hour. One server is overloaded because $4 > 3$. Two servers have maximum rate 6, so $M/M/2$ is stable. The infinite-server system is stable and has Poisson mean $4/3$. These are not contradictory predictions: they describe different available resources.

8.7 Blocking, throughput and observed arrivals

For finite capacity K, offered arrivals still occur at rate λ, but an arrival finding the system full does not enter. At stationarity, the fraction blocked is π_K under the Poisson-arrival assumptions. Accepted throughput is $\lambda(1 - \pi_K)$. In equilibrium, the long-run departure rate equals this accepted throughput, not necessarily λ. For an $M/M/1/K$ queue with $\rho \neq 1$,

$$\pi_n = \frac{(1-\rho)\rho^n}{1-\rho^{K+1}}, \qquad 0 \leq n \leq K.$$

At $\rho = 1$, the limit is uniform, $\pi_n = 1/(K+1)$. This system has a stationary law even when $\lambda \geq \mu$ because its state cannot grow beyond K; the price is a higher blocking probability.

8.8 What Little's law requires

Little's law $L = \lambda_{\text{eff}} W$ is an accounting relation between mean number in a stable system, effective arrival throughput and mean time in that system. It is not specific to exponential service. The arrival rate used must match the population counted in L. For a blocking queue, use accepted throughput rather than the offered rate. For waiting *only*, use the mean queue length L_q and mean waiting time W_q; do not mix L_q with total time in system.

8.9 Diagnosing the queue model

A real service operation may have time-varying arrivals, multiple customer classes, balking, priorities, nonexponential durations and dependent servers. Each change can break a formula without making queueing theory useless. The practical first step is to plot arrivals over time, service-duration distributions and occupancy. Then ask whether the Markov state and constant-rate assumptions are credible at the scale relevant to the decision. If they are rough approximations, sensitivity to λ and μ near the stability boundary deserves particular attention.

Exercises

Exercise 8.1. For an $M/M/1$ queue with $\lambda = 2$ and $\mu = 3$ per hour, find π_0, $\Pr(N \geq 3)$, L and W.

Exercise 8.2. For an $M/M/\infty$ system with $\lambda = 6$ and $\mu = 2$ per hour, find the stationary mean and probability that the system is empty.

Exercise 8.3. Normalise the two-machine repair model when $\lambda = \mu$ and find the stationary probability that both machines are operating.

Part III

Probability foundations

Chapter 9

Probability spaces and events

Probability starts with a precise account of which outcomes and events can be discussed. For finite experiments, every subset can be an event. On uncountable spaces, a sigma-algebra supplies a collection large enough for the calculations we need while retaining countable operations.

9.1 Outcomes and events

A sample space Ω lists possible elementary outcomes. An event is a subset of Ω. The complement A^c, union $A \cup B$ and intersection $A \cap B$ mean not A, A or B, and both, respectively. Disjoint events have empty intersection. A partition is a family of disjoint events whose union is Ω; conditioning on its members often converts one calculation into a sum of simpler ones.

Definition 9.1. A sigma-algebra $\mathcal{F}$ on Ω contains Ω, is closed under complements, and is closed under countable unions. It is consequently closed under countable intersections. A probability measure $\mathbb{P} : \mathcal{F} \to [0,1]$ satisfies $\mathbb{P}(\Omega) = 1$ and countable additivity for pairwise disjoint events:

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space.

Closure under countable operations matters for events such as infinitely many A_n occurring, written $\limsup_n A_n = \bigcap_{m \geq 1} \bigcup_{n \geq m} A_n$. This is the event in the Borel–Cantelli lemmas.

9.2 Generated and Borel sigma-algebras

The sigma-algebra generated by a collection $\mathcal{C}$, denoted $\sigma(\mathcal{C})$, is the smallest sigma-algebra containing it. For a single set A, it is $\{\varnothing, A, A^c, \Omega\}$ (with duplicates removed). For two sets A, B, their four atoms $A \cap B$, $A \cap B^c$, $A^c \cap B$, $A^c \cap B^c$ form a partition; unions of nonempty atoms make up $\sigma(A, B)$.

On $\mathbb{R}$, the Borel sigma-algebra $\mathcal{B}(\mathbb{R})$ is generated by open intervals. It includes closed and half-open intervals, countable sets and many more constructions. The phrase Borel event tells us which subsets are measurable under the chosen model; it is not a claim that every imaginable subset of $\mathbb{R}$ is Borel.

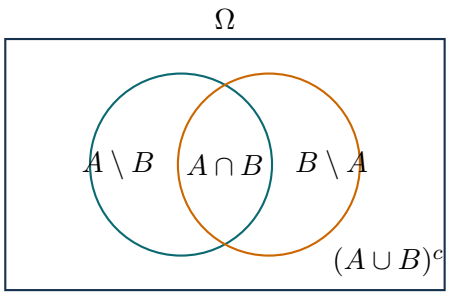


Figure 9.1: Two events partition the sample space into at most four measurable atoms.

9.3 Consequences of the axioms

Countable additivity implies $\mathbb{P}(\varnothing) = 0$, finite additivity, monotonicity, and

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

The union bound follows by replacing overlapping events with disjoint pieces:

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) \le \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

For increasing events $A_n \uparrow A$, continuity from below gives $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$. For decreasing events $A_n \downarrow A$, continuity from above gives $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$; this uses $\mathbb{P}(A_1) < \infty$, automatic for probabilities.

9.4 Borel–Cantelli: events in the long run

Theorem 9.2 (Borel–Cantelli)**.** *If $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n$ infinitely often$) = 0$. If the events A_n are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n$ infinitely often$) = 1$.*

Idea. For the first statement, $\mathbb{P}(\bigcup_{n\ge m} A_n) \le \sum_{n\ge m} \mathbb{P}(A_n) \to 0$, and the infinitely-often event lies in every tail union. For independent events, $\mathbb{P}(\bigcap_{n=m}^{N} A_n^c) = \prod_{n=m}^{N}(1 - \mathbb{P}(A_n)) \le \exp(-\sum_{n=m}^{N} \mathbb{P}(A_n)) \to 0$. Each tail almost surely contains an occurrence. □

The first direction needs no independence. Divergence of the probability sum without independence does not justify the second: the same event repeated forever is a simple counterexample.

9.5 Conditional probability and independence

When $\mathbb{P}(B) > 0$, conditional probability is $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$. If (B_i) is a finite or countable partition with positive probabilities, the law of total probability gives

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Bayes' rule follows by dividing the contribution of one cell by the total:

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

These formulas do not need a density. They are identities on events of positive conditioning probability.

Two events are independent when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Independence means observing one does not change the probability of the other, when the conditioning probabilities are positive. Pairwise independence of a collection is weaker than mutual independence. For three events, the condition for every pair does not imply $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$.

Worked example 9.3. Take two independent fair bits U, V and let $A = \{U = 1\}$, $B = \{V = 1\}$, $C = \{U \neq V\}$. Each event has probability $1/2$, and each pair is independent. Yet $A \cap B \cap C$ is impossible, so the three events are not mutually independent. This matters when using the second Borel–Cantelli lemma: its independence hypothesis concerns the family, not only pairwise correlations.

9.6 Countable operations in a concrete sequence

For events A_n, the event that at least one A_n occurs is $\bigcup_n A_n$. The event that all occur is $\bigcap_n A_n$. The event that infinitely many occur is $\limsup A_n = \bigcap_m \bigcup_{n \geq m} A_n$, while the event that all but finitely many occur is $\liminf A_n = \bigcup_m \bigcap_{n \geq m} A_n$. The first asks for an occurrence in every tail; the second asks for one tail in which every event occurs. This difference is the set-theoretic heart of many convergence arguments.

Exercises

Exercise 9.1. Let $\Omega = \{1, 2, 3, 4\}$, $A = \{1, 2\}$ and $B = \{2, 3\}$. List $\sigma(A, B)$.

Exercise 9.2. Let A_n be independent with $\mathbb{P}(A_n) = 1/n$ for $n \geq 1$. What does Borel–Cantelli say about infinitely many occurrences? What changes for $\mathbb{P}(A_n) = 1/n^2$?

Exercise 9.3. Prove continuity from above by applying continuity from below to complements.

Chapter 10

Random variables and laws

A random variable is a measurable map from outcomes to numbers. Its distribution is the probability measure it induces on the line. This distinction matters: two variables on different sample spaces can have the same distribution, while variables with the same marginal distributions can have different dependence.

10.1 Measurability and distribution

Definition 10.1. $X : \Omega \to \mathbb{R}$ is a random variable if $\{\omega : X(\omega) \le x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. Its cumulative distribution function is $F_X(x) = \mathbb{P}(X \le x)$.

It suffices to check inverse images of the generating intervals $(-\infty, x]$. The induced probability measure is $\mathbb{P}_X(B) = \mathbb{P}(X \in B)$ for Borel sets B. The CDF is nondecreasing and right-continuous, with limits 0 at $-\infty$ and 1 at $+\infty$. At a point x, its jump size is $\mathbb{P}(X = x) = F_X(x) - F_X(x-)$. A CDF may have jumps, continuous parts or both.

For a vector (X, Y), the joint CDF is $F_{X,Y}(x, y) = \mathbb{P}(X \le x, Y \le y)$. Independence means $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all Borel A, B, equivalently $F_{X,Y}(x, y) = F_X(x)F_Y(y)$ for all x, y. Knowing the two marginals is insufficient to determine the joint law.

Worked example 10.2. If U is Bernoulli$(1/2)$, let $X = U$ and $Y = 1 - U$. Both marginals are Bernoulli$(1/2)$, but $\mathbb{P}(X = 1, Y = 1) = 0$, whereas independence would give $1/4$. A different construction, $X = U$ and $Y = U$, has the same marginals but perfect positive association.

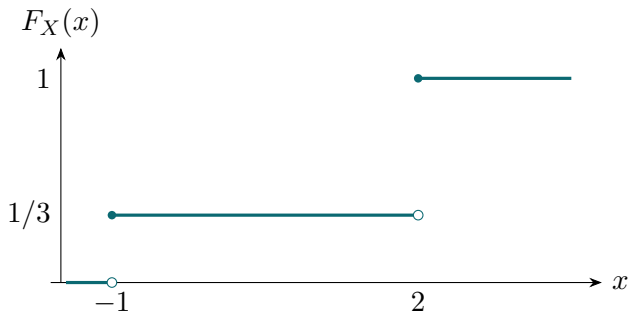


Figure 10.1: A CDF for masses $1/3$ at -1 and $2/3$ at 2. Jump sizes equal point probabilities.

10.2 Simple variables and expectation

A nonnegative simple random variable takes finitely many nonnegative values: $X = \sum_{i=1}^{m} a_i \mathbf{1}_{A_i}$ for disjoint measurable A_i, where $a_i \geq 0$. Define $\mathbb{E}X = \sum_i a_i \mathbb{P}(A_i)$. For an arbitrary nonnegative measurable X, choose increasing simple approximations $X_n \uparrow X$ and define $\mathbb{E}X = \lim_n \mathbb{E}X_n$, allowing $+\infty$. This is the Lebesgue integral.

For a real X, write $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$, so $X = X^+ - X^-$ and $|X| = X^+ + X^-$. If $\mathbb{E}|X| < \infty$, then $\mathbb{E}X = \mathbb{E}X^+ - \mathbb{E}X^-$ is finite and unambiguous. If both parts have infinite expectation, the difference of the two infinities is undefined. One can integrate with respect to the distribution:

$$\mathbb{E}g(X) = \int_\Omega g(X(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}} g(x)\, dF_X(x)$$

when the integral exists. For a pmf this becomes a sum, and for a density f it becomes $\int g(x) f(x)\, dx$.

10.3 Existence of a distribution and a process

Not every nondecreasing function is a CDF: it must also be right-continuous with limits 0 and 1. Conversely, each function with those properties defines a probability law on the Borel sets. One route is to assign an interval $(a, b]$ the mass $F(b) - F(a)$ and then extend this consistent assignment to a measure. The general Carathéodory extension theorem formalises that passage from a premeasure on a suitable algebra of sets to a measure on the generated sigma-algebra.

More generally, a consistent family of finite-dimensional distributions can define a stochastic process through the Kolmogorov extension theorem, under its standard consistency and measurable-space conditions. This theorem is an existence result; it does not by itself guarantee regular sample paths or a useful physical model.

10.4 From a CDF to probabilities

For $a < b$, $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$. The endpoints are important when X has atoms. For a continuous distribution, single points have zero probability and the choice of open or closed endpoint makes no difference. For a discrete distribution, it can change the answer. A right-continuous CDF records $\mathbb{P}(X \leq x)$, so its jump at x is exactly the atom at x.

Worked example 10.3. Let $\mathbb{P}(X = -1) = 1/3$ and $\mathbb{P}(X = 2) = 2/3$. Then $F_X(x)$ is 0 for $x < -1$, $1/3$ for $-1 \leq x < 2$, and 1 for $x \geq 2$. The probability $\mathbb{P}(-1 < X \leq 2)$ is $F_X(2) - F_X(-1) = 2/3$, while $\mathbb{P}(-1 \leq X \leq 2) = 1$.

If X has a density f, then $F_X(x) = \int_{-\infty}^{x} f(u)\, du$ and $F_X'(x) = f(x)$ at points where the derivative exists. A density may be greater than one; its integral over the whole line must equal one. A distribution can also combine atoms and a density, in which case the CDF has jumps and continuously increasing segments. The broader Lebesgue decomposition distinguishes discrete, absolutely continuous and singular continuous parts.

10.5 Transformations and joint laws

For a measurable function g, the distribution of $Y = g(X)$ is determined by inverse images:

$$\mathbb{P}(Y \in B) = \mathbb{P}(X \in g^{-1}(B)).$$

For instance, if X is Uniform$(-1, 1)$ and $Y = X^2$, then for $0 \le y \le 1$,

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = \sqrt{y}.$$

The density is $f_Y(y) = 1/(2\sqrt{y})$ on $(0, 1)$, which diverges near zero but remains integrable. Applying a one-to-one change-of-variables formula without accounting for both $+\sqrt{y}$ and $-\sqrt{y}$ would miss half the probability.

The joint law determines marginals by summing or integrating out the other coordinate. The reverse operation is not unique: fixed marginals admit many dependence structures. In the Bernoulli example above, $\mathbb{P}(X = Y)$ ranges from 0 for $Y = 1 - X$ to 1 for $Y = X$, with independent pairing giving $1/2$. A fitted marginal model therefore cannot support a claim about joint risk or simultaneous events without dependence information.

10.6 Why measurability is built into the definition

The expression $\mathbb{P}(X \le x)$ is meaningful only if the inverse-image set $\{X \le x\}$ belongs to $\mathcal{F}$. Once this holds for all x, inverse images of all Borel sets are measurable by the closure properties of sigma-algebras. The definition is minimal but powerful: sums, products, limits and continuous transformations of random variables remain measurable under standard conditions. This is what permits the expectation and convergence theorems of the next chapter to be stated without checking each event from scratch.

Exercises

Exercise 10.1. Let X take values $-2, 0, 3$ with probabilities $1/4, 1/2, 1/4$. Find F_X, $\mathbb{E}X^+$, $\mathbb{E}X^-$ and $\mathbb{E}X$.

Exercise 10.2. Construct two pairs (X, Y) with Bernoulli$(1/2)$ marginals, one independent and one dependent. Compare $\mathbb{P}(X = Y)$.

Exercise 10.3. Explain why $F(x) = 0$ for $x \le 0$ and $F(x) = 1$ for $x > 0$ is not a CDF. Change one inequality to fix it.

Chapter 11

Expectation and convergence

A sequence of random variables can converge in several ways. The distinctions are practical: a small chance of a very large value can disappear in probability while keeping the expectation large. An appropriate limit theorem must control both the type of convergence and any interchange of limit and expectation.

11.1 Inequalities and passage through a limit

Expectation is linear for integrable variables and monotone: $X \le Y$ implies $\mathbb{E}X \le \mathbb{E}Y$ when the expectations are defined. If φ is convex and X is integrable with the needed domain conditions, Jensen's inequality gives

$$\varphi(\mathbb{E}X) \le \mathbb{E}\varphi(X).$$

For $\varphi(x) = x^2$, this is $(\mathbb{E}X)^2 \le \mathbb{E}X^2$, equivalent to $\operatorname{Var}(X) \ge 0$.

Theorem 11.1 (Convergence theorems)**.** *For nonnegative $X_n \uparrow X$ almost surely, the monotone convergence theorem gives $\mathbb{E}X_n \uparrow \mathbb{E}X$, including infinite limits. For nonnegative X_n, Fatou's lemma gives $\mathbb{E}(\liminf X_n) \le \liminf \mathbb{E}X_n$. If $X_n \to X$ almost surely and $|X_n| \le Y$ for an integrable Y, dominated convergence gives $\mathbb{E}X_n \to \mathbb{E}X$.*

The hypotheses are different. Domination by an integrable *common* bound rules out probability mass escaping into increasingly tall spikes.

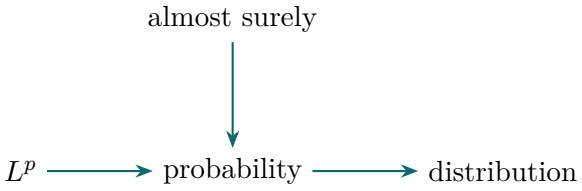
Worked example 11.2. Let $U \sim \text{Uniform}(0,1)$ and $X_n = n\mathbf{1}_{\{U \le 1/n\}}$. Then $X_n \to 0$ almost surely (except at $U = 0$, a null event), but $\mathbb{E}X_n = 1$ for every n. The spikes get rarer and taller. There is no integrable random variable dominating all X_n.

11.2 Four modes

Definition 11.3. • $X_n \to X$ almost surely if $\mathbb{P}(\{\omega : X_n(\omega) \to X(\omega)\}) = 1$.

- $X_n \to X$ in probability if $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$.
- $X_n \to X$ in L^p, $p \ge 1$, if $\mathbb{E}|X_n - X|^p \to 0$.
- $X_n \to X$ in distribution if $F_{X_n}(x) \to F_X(x)$ at every continuity point x of F_X.

The main implications are



The $L^p \Rightarrow$ probability arrow follows from Markov's inequality applied to $|X_n - X|^p$. Almost-sure convergence implies convergence in probability by continuity of probability for shrinking tail events. Convergence in probability implies convergence in distribution. The reverse arrows generally fail. When the distributional limit is a constant c, convergence in distribution to c *does* imply convergence in probability to c.

Watch the assumption

Convergence in distribution concerns laws, not samplewise closeness on a shared probability space. A variable X_n can have exactly the same distribution as X for all n without $X_n - X$ approaching zero.

11.3 Counterexamples that locate the missing assumption

Almost-sure convergence need not imply L^1: the spike sequence above converges almost surely to zero but has $\mathbb{E}|X_n| = 1$. L^1 convergence need not imply almost-sure convergence. For example, enumerate the dyadic intervals of $(0, 1]$ level by level and let X_n be the indicator of the current interval. Their lengths tend to zero, so $\mathbb{E}|X_n| \to 0$, but every point lies in one interval at each level, so the indicators take value 1 infinitely often.

The intuition is that L^1 measures average error at each n, while almost-sure convergence asks whether a *single path* eventually settles. Summability of tail probabilities is a useful bridge: if $\sum_n \mathbb{P}(|X_n - X| > \varepsilon) < \infty$ for every $\varepsilon > 0$, Borel–Cantelli yields almost-sure convergence.

11.4 Laws of large numbers

For iid integrable Y_i with $\mathbb{E}Y_i = \mu$, the strong law of large numbers states

$$\overline{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i \xrightarrow{\text{a.s.}} \mu.$$

When $\mathrm{Var}(Y_i) = \sigma^2 < \infty$, Chebyshev gives the weak law directly:

$$\mathbb{P}(|\overline{Y}_n - \mu| > \varepsilon) \le \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

This proof establishes convergence in probability, not almost-sure convergence. A stronger fourth-moment assumption gives one route to a strong law via $\mathbb{E}|\overline{Y}_n - \mu|^4 = O(n^{-2})$ and Borel–Cantelli; the general iid integrable strong law requires a deeper argument.

11.5 Characteristic functions and distributional convergence

The characteristic function $\phi_X(t) = \mathbb{E}e^{itX}$ always exists for real t. It encodes the distribution and can simplify sums of independent variables because characteristic functions multiply. Lévy's

continuity theorem says that pointwise convergence of characteristic functions to a function continuous at zero and itself a characteristic function identifies convergence in distribution. This is a route to the central limit theorem; it does not imply convergence of moments unless further control is available.

11.6 A fourth-moment route to a strong law

The weak law estimate from Chebyshev is of order $1/n$. Summing it over n diverges, so the first Borel–Cantelli lemma cannot directly turn that bound into almost-sure convergence. Under a stronger condition, $\mathbb{E}|Y_1 - \mu|^4 < \infty$, write $Z_i = Y_i - \mu$. Independence and zero means yield

$$\mathbb{E}\left(\sum_{i=1}^{n} Z_i\right)^4 = n\mathbb{E}Z_1^4 + 3n(n-1)(\mathbb{E}Z_1^2)^2 = O(n^2).$$

Dividing by n^4 gives $\mathbb{E}|\overline{Y}_n - \mu|^4 = O(n^{-2})$. Markov's inequality then gives

$$\mathbb{P}(|\overline{Y}_n - \mu| > \varepsilon) \le \frac{C}{n^2\varepsilon^4}.$$

The right-hand side is summable. Borel–Cantelli implies that for each rational $\varepsilon > 0$, only finitely many deviations exceed ε. Taking a countable intersection across those rational thresholds proves $\overline{Y}_n \to \mu$ almost surely. This argument illustrates the mechanism; the general strong law assumes only $\mathbb{E}|Y_1| < \infty$ and needs a different proof.

11.7 A comparison table

Mode	What must become small	Typical route
Almost sure	Deviations along almost every full sample path	Borel–Cantelli, strong law
Probability	Probability of a deviation over a fixed tolerance	Markov or Chebyshev
L^p	Expected pth-power error	Dominated convergence, moment bound
Distribution	CDF values at continuity points	Characteristic functions, CDF calculation

This table is a diagnostic. If a proof controls only a CDF, it has not controlled expected error. If it controls a moment, it does imply a probability statement by Markov's inequality. If it proves almost-sure convergence, it says little by itself about expectations because rare large values may remain.

11.8 Uniform integrability as the missing bridge

A family $\{X_n\}$ is uniformly integrable if

$$\lim_{K\to\infty} \sup_n \mathbb{E}\big[|X_n|\mathbf{1}_{\{|X_n|>K\}}\big] = 0.$$

Intuitively, the tails contribute negligibly to expectation uniformly in n. The spike sequence $X_n = n\mathbf{1}_{\{U\le 1/n\}}$ fails: for any K, choose $n > K$ and the displayed tail expectation is 1. When

$X_n \to X$ in probability and the family is uniformly integrable, expectations converge under the standard integrability conditions; a stronger conclusion is L^1 convergence. Dominated convergence is a convenient sufficient condition because an integrable common bound guarantees uniform integrability. This makes precise what the counterexample lacks.

Exercises

Exercise 11.1. Show with Markov's inequality that L^2 convergence implies convergence in probability.

Exercise 11.2. For $X_n = n\mathbf{1}_{\{U \le 1/n\}}$, calculate $\mathbb{P}(X_n > \varepsilon)$, $\mathbb{E}X_n$ and $\mathbb{E}X_n^2$. Identify the modes of convergence to zero that hold.

Exercise 11.3. For iid Bernoulli(p) variables, use Chebyshev's inequality to bound the probability that the sample proportion differs from p by more than ε.

Chapter 12

A first look at extremes

The final lecture briefly turns from averages to maxima. An average depends on many observations; a maximum is controlled by the upper tail. Its natural scaling can therefore differ sharply from the $\sqrt{n}$ scale of the central limit theorem.

12.1 The exact distribution of a maximum

Let $X_1, \dots, X_n$ be iid with CDF F and $M_n = \max_i X_i$. Independence gives the starting point for every calculation:

$$\mathbb{P}(M_n \le x) = \mathbb{P}(X_1 \le x, \dots, X_n \le x) = F(x)^n.$$

For any fixed x with $F(x) < 1$, this tends to zero as n grows. To obtain a nondegenerate limit, choose centring b_n and scale $a_n > 0$ so that $(M_n - b_n)/a_n$ has a meaningful limiting distribution. The choice depends on the tail of F.

12.2 Three tail behaviours

For $X_i \sim \mathrm{Exp}(\lambda)$, set $b_n = (\log n)/\lambda$ and $a_n = 1/\lambda$. Then

$$\mathbb{P}\left(\frac{M_n - b_n}{a_n} \le x\right) = \left(1 - \frac{e^{-x}}{n}\right)^n \to e^{-e^{-x}},$$

the Gumbel CDF. Exponential-like light tails lead to this family under suitable conditions.

For a Pareto tail $\mathbb{P}(X > x) = x^{-\alpha}$ for $x \ge 1$, take $a_n = n^{1/\alpha}$ and $b_n = 0$. For $x > 0$,

$$\mathbb{P}(M_n/a_n \le x) = \left(1 - \frac{x^{-\alpha}}{n}\right)^n \to e^{-x^{-\alpha}},$$

the Fréchet CDF. Heavy tails change the scaling from logarithmic to a power of n.

For Uniform$(0,1)$ observations, the upper endpoint is 1. If $Y_n = n(1 - M_n)$, then for $y \ge 0$,

$$\mathbb{P}(Y_n > y) = \mathbb{P}(M_n < 1 - y/n) = (1 - y/n)^n \to e^{-y}.$$

Equivalently, $n(M_n - 1)$ has a reverse-exponential limit supported on the nonpositive line, an example of the Weibull extreme-value type. A finite endpoint changes both centring and support.



Figure 12.1: The three extreme-value types correspond to different upper-tail structures.

12.3 What the classification does and does not say

The classical extreme-value theorem states that any nondegenerate limit of suitably affine-normalised iid maxima belongs to one of three types, under its conditions. It does not claim every distribution admits such a limit, and it does not identify a tail family from a short sample. In applications, dependence, changing distributions, measurement limits and threshold choice can dominate the theoretical asymptotic approximation. The lecture introduces this landscape; a full course in extreme-value inference would require additional material.

12.4 Reading the tail from the scaling

The order of the maximum can often be guessed by asking for a level u_n so that one exceedance is expected among n observations:

$$n\mathbb{P}(X > u_n) \approx 1.$$

For an exponential tail $e^{-\lambda u}$, this gives $u_n \approx (\log n)/\lambda$. For a Pareto tail $u^{-\alpha}$, it gives $u_n \approx n^{1/\alpha}$. For a bounded distribution with endpoint x_F, it gives a level approaching x_F from below. This heuristic suggests a normalisation, but the exact limit still requires calculating $F(u_n)^n$.

The same tail reasoning explains why ordinary averages and maxima answer different risk questions. A finite-variance central limit theorem describes fluctuations near the mean. It says little about an extreme quantile controlled by a heavy tail. Conversely, an extreme-value model focuses on the upper tail and should not be used to infer a mean without checking whether that mean exists.

12.5 Dependence changes the effective sample

The identity $\mathbb{P}(M_n \le x) = F(x)^n$ relies on independence. If all X_i are exactly the same variable, then $M_n = X_1$ and there is no growing maximum at all. Weak dependence can still allow classical extreme-value limits, often with a correction describing clusters of exceedances, but such conditions are not automatic. When measurements are taken close together in time or space, treating them as iid can exaggerate the effective number of independent opportunities for an extreme.

Exercises

Exercise 12.1. For n iid Uniform$(0, 1)$ variables, find $\mathbb{E}M_n$ and verify that it tends to the upper endpoint.

Exercise 12.2. For iid Exp(2) variables, identify a_n, b_n that yield the Gumbel limit.

Exercise 12.3. A model has Pareto upper tail with exponent $\alpha = 2$. By what factor does the typical maximum scale when sample size is multiplied by 4?

Appendix A

Solutions to selected exercises

The solutions are deliberately concise, but they show the step that determines the answer. A numerical result without its conditioning or normalisation argument is rarely enough in this course.

Chapter 1: Processes

1.1. Let $B \sim \text{Bernoulli}(p)$ and set $X_n = B$ for all n. Let Y_n be independent Bernoulli(p) variables. Each marginal is Bernoulli(p), but $\mathbb{P}(X_1 = X_2) = 1$ and $\mathbb{P}(Y_1 = Y_2) = p^2 + (1-p)^2$.

1.2. Stationary increments specify the law of $N(t+h) - N(t)$ for a given length h, not a common law for levels. A rate-λ Poisson process has stationary increments, while $N(t) \sim \text{Poisson}(\lambda t)$ varies with t.

Chapter 2: Branching

2.1. $m = 0(2/5) + 1(1/5) + 2(2/5) = 1$. The pgf is $f(s) = 2/5 + s/5 + 2s^2/5$. Solving $f(s) = s$ gives $(s-1)^2 = 0$, so extinction has probability 1. The critical law is nondegenerate and has $p_0 > 0$.

2.2. Each ancestor starts an independent family. All r families must die out, giving q^r.

2.3. Conditional on Z_n, the next mean and variance are mZ_n and $\sigma^2 Z_n$. Total variance gives $v_{n+1} = \sigma^2 m^n + m^2 v_n$. At $m = 1$ this reduces to $v_{n+1} = v_n + \sigma^2$, hence $v_n = n\sigma^2$. The expression for $m \neq 1$ has this same limit because $(m^n - 1)/(m-1) \to n$.

Chapter 3: Random walks

3.1. The success probability is $5/8$ and the expected time is $5(8-5) = 15$ steps.

3.2. Here $q/p = 1/2$, so $h_1 = (1 - 1/2)/(1 - (1/2)^4) = 8/15$, above the fair value $1/4$.

3.3. For $e_x = x(d-x)$, $(e_{x+1} + e_{x-1})/2 = e_x - 1$, so $e_x = 1 + (e_{x+1} + e_{x-1})/2$. Also $e_0 = e_d = 0$.

Chapter 4: Markov basics

4.1. $(\boldsymbol{P}^2)_{12} = .9(.1) + .1(.6) = .15$. Balance $.1\pi_1 = .4\pi_2$ and normalisation give $(\pi_1, \pi_2) = (.8, .2)$.

4.2. Uniform $(1/3, 1/3, 1/3)$ is stationary. Starting at 1, the return probability is 1 when n is divisible by 3 and 0 otherwise, so it has no limit.

4.3. For m states, $(\boldsymbol{\pi}\boldsymbol{P})_j = m^{-1}\sum_i p_{ij} = m^{-1} = \pi_j$ because column j sums to one.

Chapter 5: Long-run Markov behaviour

5.1. The classes are $\{1, 2\}$ and $\{3\}$; only $\{3\}$ is closed. States 1 and 2 are transient, and the sole stationary distribution is $(0, 0, 1)$.

5.2. Vertex degrees on the four-vertex path are $(1, 2, 2, 1)$, so $\boldsymbol{\pi} = (1, 2, 2, 1)/6$. The graph is bipartite, the walk has period 2, and $\boldsymbol{P}^n$ does not converge.

5.3. Local balance gives $\pi_0 = \pi_1/3$ and $(2/3)\pi_1 = \pi_2$. Thus the weights are $(1, 3, 2)$ and $\boldsymbol{\pi} = (1, 3, 2)/6$.

Chapter 6: Poisson processes

6.1. In half an hour the mean count is $3/2$, so $\mathbb{P}(N(.5) = 2) = e^{-3/2}(3/2)^2/2$. Twenty minutes is $1/3$ hour, so $\mathbb{P}(T_1 > 1/3) = e^{-1}$.

6.2. Conditional on $N(2) = 3$, the three unordered event times are independent Uniform$(0, 2)$. Each is in the first hour with probability $1/2$, giving $1/8$.

6.3. Solve $1 - e^{-2\pi r^2} = 1/2$: $r = \sqrt{\log 2/(2\pi)}$ kilometres.

Chapter 7: Continuous-time Markov chains

7.1. The stationary law is $(3/5, 2/5)$ and $\mathbb{P}(X(t) = 1 \mid X(0) = 0) = (2/5)(1 - e^{-5t})$.

7.2. The generator is $\boldsymbol{Q} = \begin{pmatrix} -2 & 2 & 0 \\ 1 & -4 & 3 \\ 0 & 4 & -4 \end{pmatrix}$. The nonzero jump probabilities are $k_{01} = 1$, $k_{10} = 1/4$, $k_{12} = 3/4$, and $k_{21} = 1$.

7.3. Off-diagonal rates are nonnegative. Setting q_{ii} to minus their sum makes the row sum zero and represents the rate of leaving state i.

Chapter 8: Queues

8.1. $\rho = 2/3$, so $\pi_0 = 1/3$, $\mathbb{P}(N \geq 3) = \rho^3 = 8/27$, $L = \rho/(1 - \rho) = 2$, and $W = L/\lambda = 1$ hour.

8.2. Stationary N is Poisson with mean $\lambda/\mu = 3$; $\mathbb{P}(N = 0) = e^{-3}$.

8.3. When $\lambda = \mu$, the weights are $(1, 2, 2)$ and normalise to $(1/5, 2/5, 2/5)$. Both machines operate in state 0, with probability $1/5$.

Chapter 9: Probability spaces

9.1. The four atoms are $\{1\}, \{2\}, \{3\}, \{4\}$, so every subset is generated: $\sigma(A, B) = 2^{\Omega}$.

9.2. The harmonic series diverges, so independent A_n with probabilities $1/n$ occur infinitely often almost surely. The sum of $1/n^2$ converges, so those events occur only finitely often almost surely; this direction does not require independence.

9.3. If $A_n \downarrow A$, then $A_n^c \uparrow A^c$. By continuity from below, $\mathbb{P}(A_n^c) \uparrow \mathbb{P}(A^c)$, hence $\mathbb{P}(A_n) = 1 - \mathbb{P}(A_n^c) \downarrow \mathbb{P}(A)$.

Chapter 10: Random variables

10.1. $F_X(x)$ is 0 for $x < -2$, $1/4$ for $-2 \le x < 0$, $3/4$ for $0 \le x < 3$, and 1 for $x \ge 3$. $\mathbb{E}X^+ = 3/4$, $\mathbb{E}X^- = 1/2$, so $\mathbb{E}X = 1/4$.

10.2. Independent fair Bernoulli variables agree with probability $1/2$. Setting $Y = X$ gives the same marginals but agreement probability 1.

10.3. The proposed F is not right-continuous at zero: $F(0) = 0$ but $F(0+) = 1$. Set $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x \ge 0$.

Chapter 11: Convergence

11.1. Markov applied to $|X_n - X|^2$ gives $\mathbb{P}(|X_n - X| > \varepsilon) \le \mathbb{E}|X_n - X|^2/\varepsilon^2 \to 0$.

11.2. For fixed $\varepsilon > 0$ and $n > \varepsilon$, $\mathbb{P}(X_n > \varepsilon) = 1/n \to 0$. Yet $\mathbb{E}X_n = 1$ and $\mathbb{E}X_n^2 = n$. Thus $X_n \to 0$ in probability and almost surely, but not in L^1 or L^2.

11.3. The sample proportion has mean p and variance $p(1-p)/n$. Therefore $\mathbb{P}(|\overline{Y}_n - p| > \varepsilon) \le p(1-p)/(n\varepsilon^2)$.

Chapter 12: Extremes

12.1. $F_{M_n}(x) = x^n$ for $0 \le x \le 1$, so the density is nx^{n-1} and $\mathbb{E}M_n = n/(n+1) \to 1$.

12.2. Use $b_n = (\log n)/2$ and $a_n = 1/2$.

12.3. The maximum's scale is $n^{1/2}$, so multiplying n by 4 multiplies this scale by 2.

Appendix B

Lecture coverage map

This map records the primary destination of each supplied lecture set. Overlapping material is cross-referenced in the prose. The title of Lecture 23's file includes a parenthesised suffix in the source folder.

Lecture	Material read	Main destination
01	Course outline; stochastic-process and probability themes	Chapters 1, 9–12
02	Process definitions, finite-dimensional distributions, branching pgf	Chapters 1–2
03	Branching moments and extinction	Chapter 2
04	Extinction cases; random-walk setup, LLN and CLT	Chapters 2–3, 11
05	Symmetric gambler's ruin hitting probability	Chapter 3
06	Ruin duration; Markov property and transition steps	Chapters 3–4
07	Transition matrices, Chapman–Kolmogorov, spectral view	Chapter 4
08	Stationarity, doubly stochastic chains, examples	Chapters 4–5
09	Stationary versus limiting laws; recurrence, periodicity	Chapter 5
10	Communication, closed classes, absorbing states and period	Chapter 5
11	Global and detailed balance; graph walks, birth–death chains	Chapter 5
12	Time reversal, cycle criterion, recurrence; Poisson prelude	Chapters 5–6
13	Poisson counts, exponential waits, memorylessness	Chapter 6
14	Poisson equations, rare-event limit, ordered arrival times	Chapter 6
15	Nonhomogeneous and spatial Poisson processes	Chapter 6
16	Spatial voids, superposition, thinning and simulation	Chapter 6
17	CTMC definition, jump chain, holding times, regularity	Chapter 7
18	Queue rates, generator, forward and backward equations	Chapters 7–8
19	CTMC limits, stationarity, recurrence and two-state models	Chapter 7
20	Repair model, birth–death and queue families	Chapter 8
21	Local balance; $M/M/1$, $M/M/\infty$, $M/M/c$ stability	Chapter 8
22	Sample spaces, event operations, partitions	Chapter 9
23	Sigma-algebras and probability spaces	Chapter 9

Lecture	Material read	Main destination
24	Generated and Borel sigma-algebras; axioms and continuity	Chapter 9
25	Discrete and continuous examples; Borel–Cantelli	Chapter 9
26	Measurable random variables, CDFs, vectors and independence	Chapter 10
27	Dependence with fixed marginals; simple variables	Chapter 10
28	Extension and CDF theorems; Lebesgue expectation	Chapter 10
29	Jensen, monotone convergence, Fatou and domination	Chapter 11
30	Convergence modes, implication diagram, laws of large numbers	Chapter 11
31	Counterexamples to reverse implications; strong LLN route; maxima	Chapters 11–12

Using the models

The local rules in Parts I and II yield global predictions only under their stated assumptions. Part III makes the probabilistic meaning of those predictions precise. In an application, identify the state, clock, independence assumptions and target quantity before treating any formula as an answer.